

Flying STAR, a Hybrid Crawling and Flying Sprawl Tuned Robot

Nir Meiri, and David Zarrouk

Abstract—This paper presents Flying STAR (FSTAR) a reconfigurable hybrid flying quadcopter robot. FSTAR is the latest in the family of the STAR robots fitted with a sprawling mechanism and propellers allowing it to both run and fly using the same motors. The combined capabilities of running and flying allows FSTAR to fly over obstacles or run underneath them and move inside pipes. The robot can reduce its width to crawl in confined spaces or underneath obstacles while touching the ground. We first describe the design of the robot and the configuration of the wheels and propellers in the flying and running modes. Then we present the 3D printed prototype of the FSTAR robot which we used for our experiments. We evaluate the energy requirements of the robot and the forces it can generate. The experimental robot can fly like an ordinary quadcopter but can also run on the ground at a speed of up to 2.6 m/s to save energy (see video).

Index Terms— Crawling Robot, Flying Robot, Mechanical Design, Reconfigurable Robot, Sprawl Tuning.

I. INTRODUCTION

Miniature crawling robots have long been developed for moving inside unstructured environments for search and rescue, excavation and reconnaissance missions. Their small size, low weight and high navigability enable their deployment in large numbers to quickly inspect large areas. Much effort has been invested in reducing their size, increasing their speed and lowering their energy consumption. Examples of crawling robots include Mini-Whegs [1], Dyna-RoACH [2], DASH [3], iSprawl [4], OctoRoACH [5], RHex [6], and Sprawlita [7] STAR [8], 1STAR [9] TAYLRoACH [10] and RSTAR [11]. Many substantial advances have been made in the design of these robots using passive mechanics, minimal actuation [12]–[14] and reconfigurable mechanics [15]–[22]. One of their limitations, however, remains that the size of obstacles they can overcome is proportional to their body and leg sizes. Small flying robots, on the other hand, for example [23]–[28], can easily fly over large obstacles. However, their travelling capabilities are limited inside confined spaces where their wings or propellers can collide with walls or other challenging environments and dust or debris can interfere with their flight and sensory performance. To overcome this limitation, some hybrid robots that can combine flying and running capabilities have been developed [29]–[30]. Most of them exhibit better flying capabilities than ground running performance.

In previous works [8]–[11], we presented two sprawl-tuned robots (STAR and its upgrade RSTAR) which can actively adjust their sprawl angles to transform their dynamics between the lateral and the sagittal planes through the use of a variable sprawl angle. The RSTAR fitted with

an extra four bar mechanism exhibited superior performance (compared to STAR) in moving on variety of surfaces and traversing obstacles.



Figure 1. The FSTAR is a newly developed robot that can sprawl its legs up and down. The robot can act like a regular flying quad when the sprawl is zero (bottom view) and like a car when the sprawl is down and the wheels are touching the ground (bottom).

The new hybrid flying robot FSTAR presented in this paper (Figure 1) belongs to the family of STAR robots [8] [9] [14] [11]; i.e., it can also vary its sprawl angle. FSTAR features unique capabilities in that it can transform itself from a flying quadcopter into a running robot using its sprawling mechanism. The total length of the robot is 24.6 cm and its width in the flying mode is 37.1 cm. The total weight of the robot is 900 grams including the battery and control board.

This paper is organized as follows. We first present the mechanical design of the robot in Section II. In section III, we present the kinematic analysis of the sprawl angle and the dynamics of crawling over the ground. In Section IV, we present the experiments we performed on the robot to determine its energy consumption and its capabilities in both the crawling and flying modes.

II. DESIGN AND MANUFACTURING

The primary design goals of the FSTAR were to achieve high performance in both crawling and flying. Its weight was lowered to increase its maneuverability and its payload of sensors needed for search and rescue missions. In parallel, the robot had to be energy efficient so as to increase its work range and work time without recharging.

A. Robot Design

FSTAR has a rigid body core which houses the

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controllers, the onboard batteries, and the sprawl mechanism which is actuated by a servo motor. The four motors that simultaneously actuate the propellers and the wheels are tilted together with the sprawl mechanism. Both sides of the sprawling mechanism are phased together and move symmetrically relative to its center body. When in contact with the ground, the back wheels provide propulsion during running whereas the forward wheels are passive.

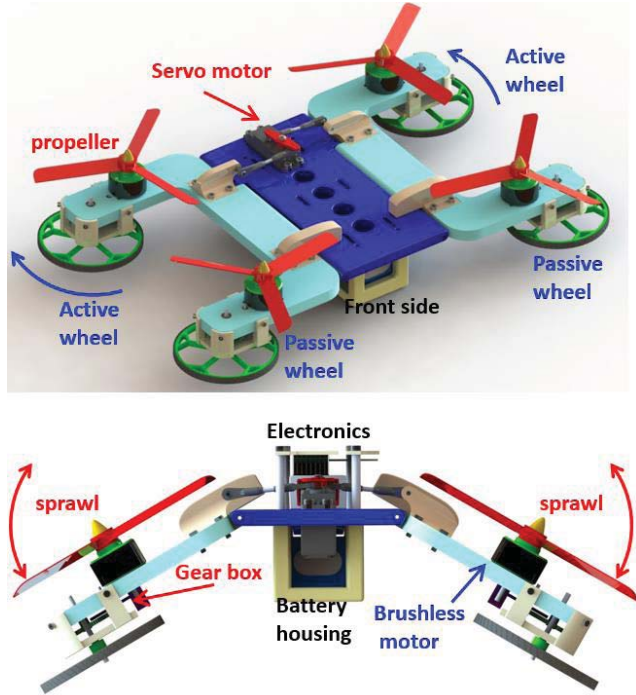


Figure 2. The mechanical design of the FSTAR robot. Isometric view in its flight configuration (top) and back view (bottom) of the robot in its crawling configuration. The robot has two active back wheels and two passive front wheels.

1) The Sprawling Mechanism

The sprawl angle ρ is the relative angle between the plane of the wheels (and propellers) and the main body, as presented in Figure 2. The sprawl angle is defined as $\rho=0$ when the plane of the wheels is parallel to the body, and the positive sense of the sprawl angle is downwards.

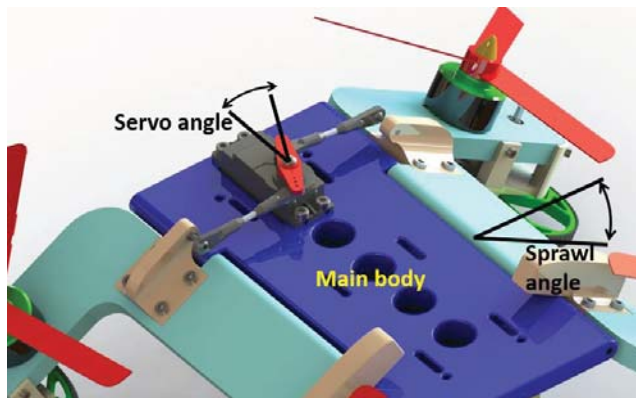


Figure 3. The sprawl mechanism, actuated using a servo motor, has a range of 55 degrees.

The sprawl angles at both sides are actuated symmetrically through the servo motor to insure an identical sprawl on both sides of the robot. The sprawl angle in this design can be varied in the range of 0 to 55 degrees as shown in Figure 2. The positive sense of the sprawl angle is defined downwards.

2) The motor housing and gear boxes.

The propellers of the robot are directly attached to the axis of the motors. The front wheels are passive (unactuated) whereas the back wheels are actuated using the same motors of the quadcopter via a 20:1 spur gearbox which we designed to reduce the speed of the wheels and increase the torque. The radius of the wheels is 3.85 cm and we used 12.7 cm (5 inches) long, 10.2 cm pitch (4 inches) three-blade propellers.

B. Actuation and Control

The FSTAR is actuated with four brushless motors for flying and one servo motor that actuates the sprawl mechanism. In this design, we used racing-drone brushless motors (MultiStar 2204-2300KV). The motors weigh 26 grams and can generate 120 Watts of power each, according to their catalog (see our own experiments on the motors in section IV). The servo motor (JR DS8717) with a rotational range of 120 degrees, weighs 69 grams and produces a torque of 140 Ncm. We used an off-the-shelf (DJI NAZA Multirotor Lite) controller fitted with three axis gyroscopes and three axis accelerometers. The speed controller is the Afro 12A UltraLite Multi-rotor ESC V3 OPTO which can provide 12A of current.

C. Manufacturing

Most of the mechanical parts of the body of the FSTAR were manufactured using in-house 3D printing. We used an Objet Connex 350 3D printer whose accuracy is roughly 0.05 mm (Stereolithography - material type VeroWhite RGD 835) and UpBox (Fuse Deposition Modeling - ABS) whose accuracy is roughly (0.2mm). We invested considerable effort in simplifying part replacement, which is essential when performing risky maneuvers with the robot (although little maintenance was required during the experiments).

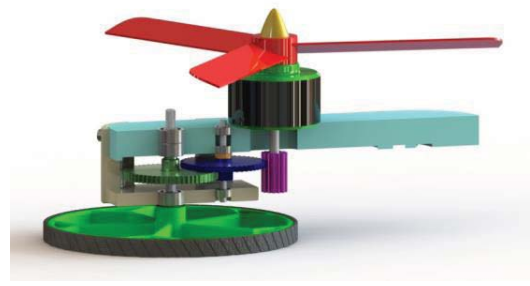


Figure 4. The active wheels motor housing and gearboxes with 20:1 gear ratio.

III. KINEMATIC AND DYNAMIC ANALYSIS

In this section, we analyze the kinematics and dynamics of the robot. We present the different configurations that the robot can achieve in both crawling and flying and evaluate the gear ratio required to drive the robot and transit between flying and running. The results were then implemented in the actual design of the flying STAR robot.

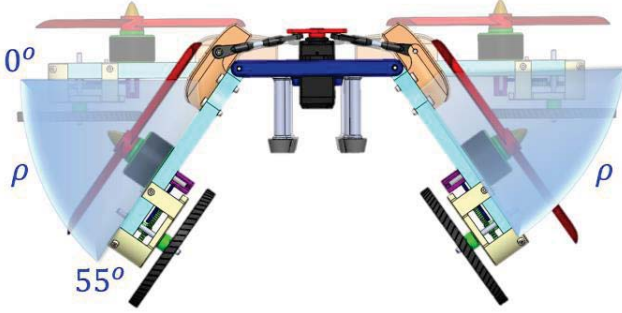


Figure 5. The work volume of the sprawling wheels and propellers of the FSTAR. The sprawl range of the current design is 55 degrees.

A. Kinematic Analysis

1) Sprawling

Similar to the other STAR robots, the FSTAR has a sprawl mechanism that changes its width and height. At the maximum sprawl angle of 55 degrees, the mechanism is mechanically locked; i.e., the servo motor does not have to produce any torque to keep the sprawl at the maximum angle (see Figure 6).

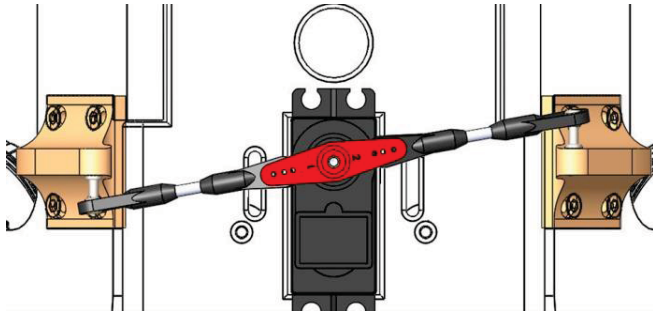


Figure 6. The sprawl mechanism is locked when the sprawl is at its maximum angle.

To determine the relationship between the servo motor angle and the sprawl angle, we used the “motion study tool kit” of Solidworks program which provided the direct relationship between the two angles. The relationship between the two angles is presented in Figure 7. The maximum derivative of the graph representing the sprawl angle as a function of the servo angle is obtained at zero sprawl with a value of 0.78:

$$\Delta\theta_{\text{servo}} = 0.78\Delta\rho \quad (1).$$

At the maximum sprawl angle, the derivative becomes zero, implying that the mechanism is in the “locking stage” and the servo motor does not have to produce any torque to maintain the sprawl angle at its maximum value.

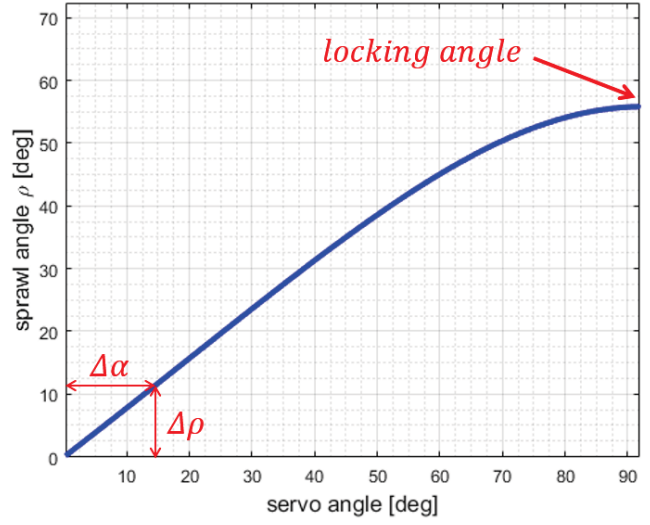


Figure 7. The relationship between the servo motor angle and the sprawl angle ρ as obtained by the “motion study tool kit” of Solidworks program.

2) Running

The running speed of the robot is a function of the angular speed of the motor, where the gear ratio of 20:1 is expressed as:

$$V_{\text{running}} = \frac{\dot{\theta}_{\text{motor}}}{20} R_{\text{wheels}} \quad (2).$$

For a wheel size of 3.85cm, the running speed of 1m/s corresponds to nearly 5000 RPM (exactly 4960RPM). Note that at nearly 14000 RPM, the robot becomes airborne in its zero sprawl configuration (see section IV) which limits the speed to 2.82m/s at low sprawl angles.

B. Force and Torque Analysis

In this section, we calculate the forces acting on the robot and the torques that must be provided by the different motors of the robot when running over a horizontal surface and in flight mode. For the sprawling mode, we analyze the cases which require larger torques; i.e., the robot increases its sprawl and therefore lifts its center of mass (COM). We denote by F_{normal} , F_{side} , the absolute values of the normal and side forces acting on one side of the legs, (see Figure 8).

The normal force F_{normal} acting on the legs (assuming low acceleration) is:

$$|F_{\text{normal}}| = \frac{mg}{2} - 2F_{\text{lift}} \cdot \cos(\rho) \quad (3).$$

Where m is the total mass of the robot. The side force F_{side} when the robot is extending its sprawl is:

$$|F_{\text{side}}| = \mu \left(\frac{mg}{2} - 2F_{\text{lift}} \cdot \cos(\rho) \right) \quad (4).$$

Note that F_{side} is pointed outwards when the robot increases the sprawl angle and inwards when it decreases the sprawl.

A force diagram of the robot is presented in Figure 8. The torque acting on the sprawl joint T_{sprawl} is a function of the sprawl angle ρ and whether or not the propellers are

providing lift:

$$|T_{sprawl}| = \left(\frac{mg}{2} - 2F_{lift} \cdot \cos(\rho) \right) \times (L_1 \cos(\rho) + \mu(L_1 \sin(\rho) + L_2 \cos(\rho))) \quad (5).$$

Where F_{lift} is the lifting force generated by a single propeller. The lengths $L_1=10.3$ cm and $L_2=2.9$ cm are defined respectively as the distances from the sprawl joint to the ground in parallel and vertical to the sprawl arm (see Figure 8). The maximum torque required by the servo motor is when the sprawl is almost zero [11]. By inserting Eq. (1) we obtain:

$$|T_{servo}| = 2 \times 0.78 \cdot \left(\frac{mg}{2} - 2F_{lift} \cdot \cos(\rho) \right) \times (L_1 \cos(\rho) + \mu(L_1 \sin(\rho) + L_2 \cos(\rho))) \quad (6).$$

The torque required by the servo motor is clearly at its maximum when the lifting force is zero. In this case, assuming that the coefficient of friction (COF) μ is 0.3 (plastic contact with a tile floor), the maximum torque generated by the servo T_{servo} is 78.4 Ncm (nearly half of the torque which the servo motor can produce).

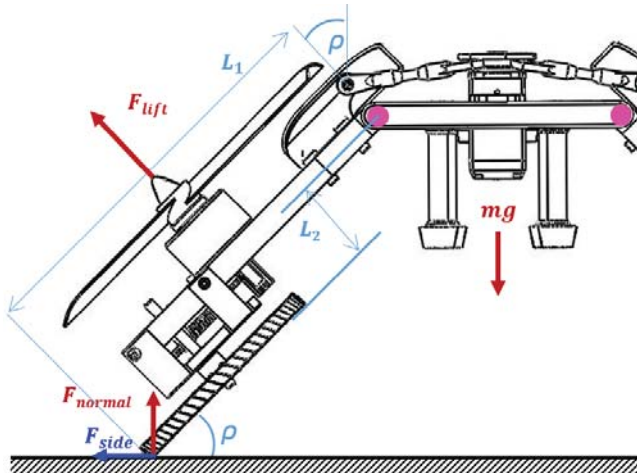


Figure 8. The forces acting on the robot during sprawling up. The normal force and the friction side forces resist the motion.

The torque generated by the wheels is:

$$T_{wheels} = 20 \cdot (T_{motor} - T_{propeller}) \quad (7).$$

where T_{motor} is the torque that the motor can generate and the $T_{propeller}$ is the torque required to rotate the propellers. Therefore the combined thrust force that the two active wheels can generate, assuming sticking to the surface, is:

$$F_{thrust} = 2 \cdot \frac{20 \cdot (T_{motor} - T_{propeller})}{R_{wheels}} \quad (8).$$

At speeds below 1m/s, the torque of the propellers $T_{propellers}$ can be neglected (see Figure 11). Therefore, based on our parameters and assuming $T_{motor}=0.05$ Nm, the thrust force is 52 N.

As the robot increases its speed, the lift generated by the propellers increases as a function of the speed squared (see Figure 11). If the total lift force exceeds the weight, the robot will start flying. The total lift force as a function of the sprawl angle is:

$$F_{lift_tot} = 4F_{lift} \cos(\rho) \quad (9).$$

Since the weight of the robot is nearly 9N, the required lift force at a zero sprawl angle by each motor is 2.25 N, which occurs at nearly 14000 RPM. By Eq. (2), this speed corresponds to nearly 2.82 m/s running. Inversely, the maximum sprawl angle which will still allow flying can also be calculated using:

$$\rho = \arccos\left(\frac{mg}{4 \max(F_{lift})}\right) \quad (10).$$

The maximum lift force using the four motors is obtained at 18000 RPM for a value of 12.4 N. By inserting into Eq. (10), we obtain a maximum sprawl angle of 43.4 degrees.

IV. RESULTS

This section presents the results of the experiments conducted on the robot. We tested the FSTAR during running and flying and evaluated its energy consumption.

A. On Ground Running

We performed multiple runs with the onboard flight controller and found that the controller in a closed loop with the gyros was able to keep the robot in a straight line. The robot was run at speeds of nearly 50 cm/s and a low sprawl angle of 15 degrees.

Figure 9 presents the results of 10 runs (out of 10) that were recorded using our Optitrack setup (12 Prime 13 cameras – 1.3 MP) autonomously and with a human operator. In the first experiment, the robot ran in a nearly straight line after an initial average deviation of 5.7 degrees. Note that the error appeared in its first accelerating stage and was partially due to orientation inaccuracy when placing the robot at the start of the experiment. Figure 9 presents the results of the robot running autonomously in a straight line after an initial deviation and the robot being run by a human operator.

B. On Ground Turning Radius

In our second experiment we rotated the robot in its nearly flat configuration (almost zero sprawl angle) and at its maximum sprawl angle (55°). The turning results can be seen in Figure 10. The turning radii for the low sprawl and high sprawl angles were respectively 28 cm and 27 cm. Note

that in theory, the robot can make a full rotation in place (with zero turning radius) if it can rotate its wheels backwards. However, turning the wheels backwards is currently impossible due to the electronic limitations of the control system.

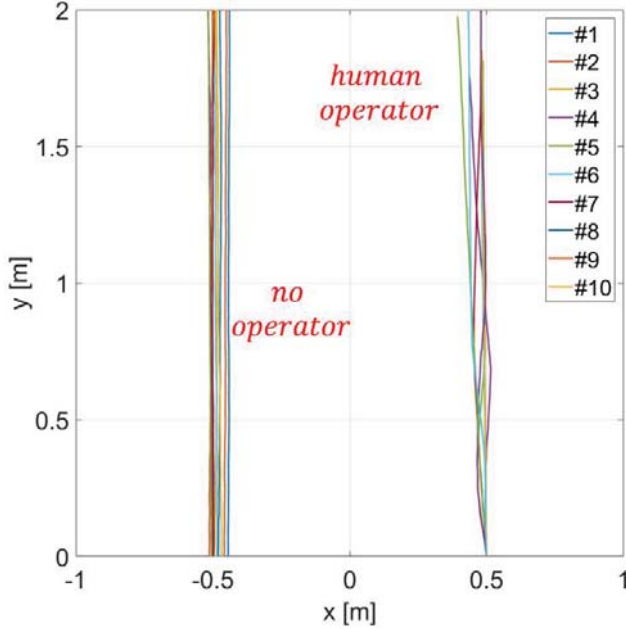


Figure 9. FSTAR path in controlled running, showing the path of 10 experiments as recorded using the Optitrack tracking system. After an initial correction, the controller drives the robot in a nearly straight line.

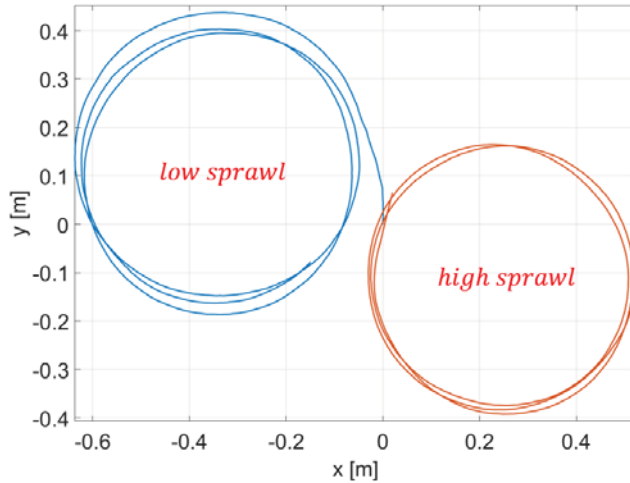


Figure 10. The turning radius of the robot for both high sprawl and low sprawl as recorded by the Optitrack setup.

C. Power and Thrust Experiments

To estimate the lift force as a function of the motor speed and the power consumption of the motors with and without the wheels, we used a “RC Benchmark 1580” dynamometer which experimentally measured the electric power consumption, thrust and torque produced by the motors (see Figure 11). The thrust experiment indicated that the robot could take off at nearly 14000 RPM (at zero sprawl by

generating a full force of 9 Newtons). The speed of the wheels at this stage was (using Eq. (2)) 2.82 m/s which is comparable to what we observed during our experiments. At flying speeds (exceeding 14000 RPM), the power consumption of the motor with the wheels increased by less than 5% as compared to the case of flight without the wheels. Given the fact that only two wheels are actuated, the total increase in energy consumption was only 2.5%. Note that the power consumption also increases because of the extra weight the robot is carrying. Each of the motors can generate 0.05 Nm of torque. When multiplying by the gear ratio and dividing by the radius of the wheels using Eq. (8), the thrust running force that each motor can generate is 26 N. Since two wheels are attached to the motors, the total running force is 52 N, which is substantially greater than the weight of the robot and sufficient to drive it in nearly all conditions.

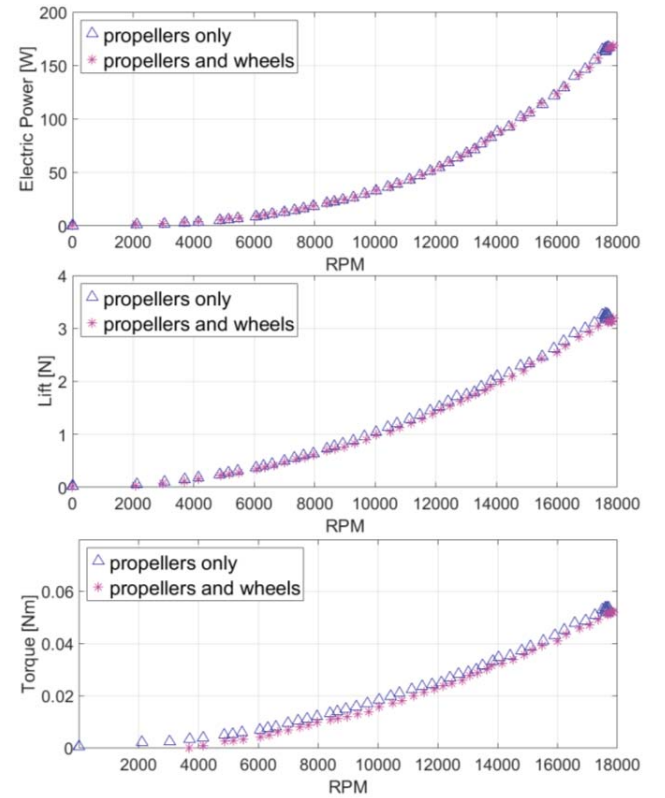


Figure 11. Mechanical experiments performed on a single motor fitted with propellers both with and without wheels to determine its thrust force, torque and power requirements in both configurations.

D. Combined Flying and Running Experiments

We experimentally validated the robot by flying it and running it over the ground (see Figure 12 and video). The robot in flying mode in (a) lands in (b) and re-orientates itself (c) to enter underneath an obstacle (d) and exit it (e). After that the robot sprawls up (f) to go between two obstacles and transit to flying again (g-h). The experiment shows that can smoothly transit between its flying and driving modes to go underneath obstacles, squeeze into tight spaces and continue flying afterwards (see video).

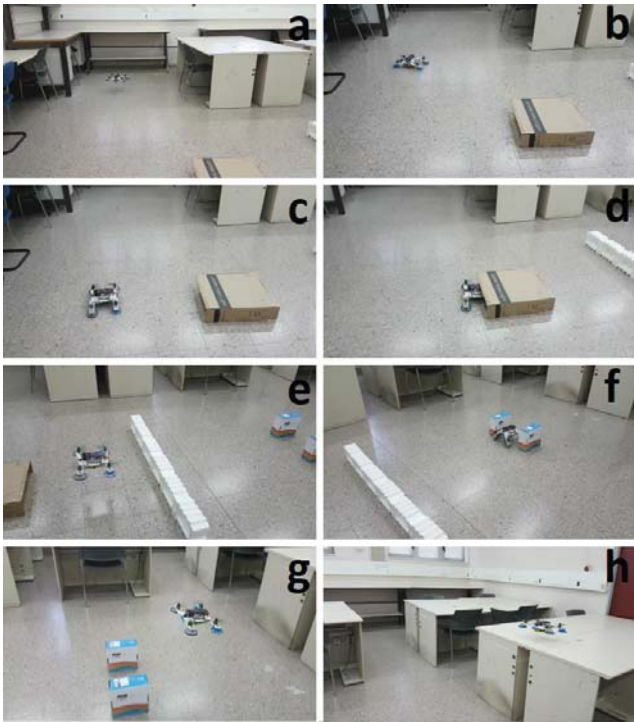


Figure 12. FSTAR performing combined flying and crawling experiments outdoor (see video).

E. Running over a Variety of Surfaces

Finally, we tested the robot outdoors on a variety of surfaces (see Figure 13 and video). The robot can crawl over grass and the ground. The highest ground speed we recorded for the robot was 2.6 m/s at 55 degrees sprawl angle and 2.21 m/s at 15 degrees sprawl. Beyond that speed the robot becomes partially (two wheels disconnected from the ground) or fully airborne.

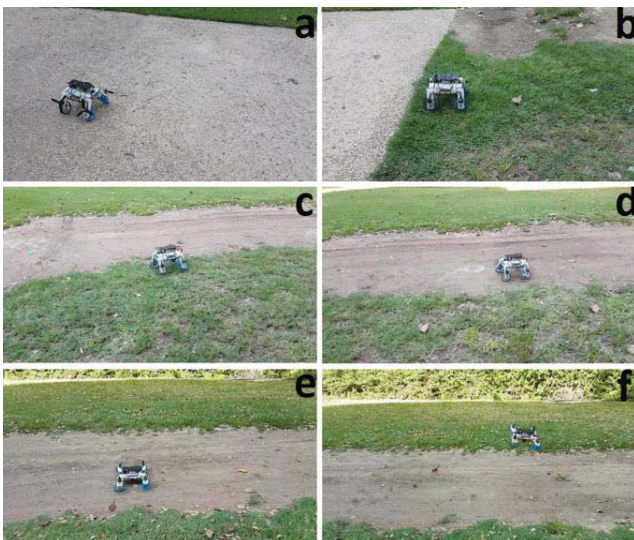


Figure 13. FSTAR crawling over concrete, grass and ground and then flying (see video).

V. CONCLUSION

In this paper we presented our novel flying sprawl- tuned robot that can travel on the ground and fly using the same

motors. The robot is reconfigurable and is fitted with both wheels and propellers. The sprawl, which can be varied from a flat configuration to 55 degrees allows the robot to transform its kinematics from a flying quadcopter to a car like-robot. The FSTAR can perform multiple unique maneuvers combining flying and running in open and confined spaces.

We conducted a dynamic analysis to evaluate the design parameters and constraints used to manufacture the robot. We found that the addition of the wheels for running increased the flight cost of transport by less than 2.5%, making the robot energy efficient. The robot proved to be highly maneuverable and the transition between the flying and running modes is very smooth. The robot can rotate on the ground with a turning radius of 28 cm and run at speeds of more than 2.5 m/s before becoming airborne. Throughout our experiments, we found that the robot is highly reliable and that practically no maintenance was required.

Our future goals are to develop a locomotion planning algorithm for the robot which combines both crawling and flying strategies to optimally exploit its capabilities in terms of speed and cost of transport. We also plan to implement a clutch system that can disengage the actuation of the propellers from the wheels.

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